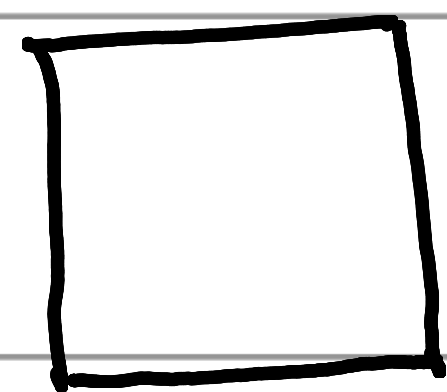


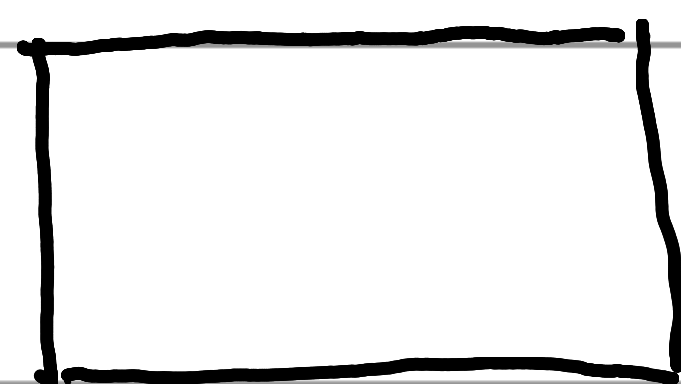
23 Sep' 25 // 28 Sep' 26

Geometry — Shapes



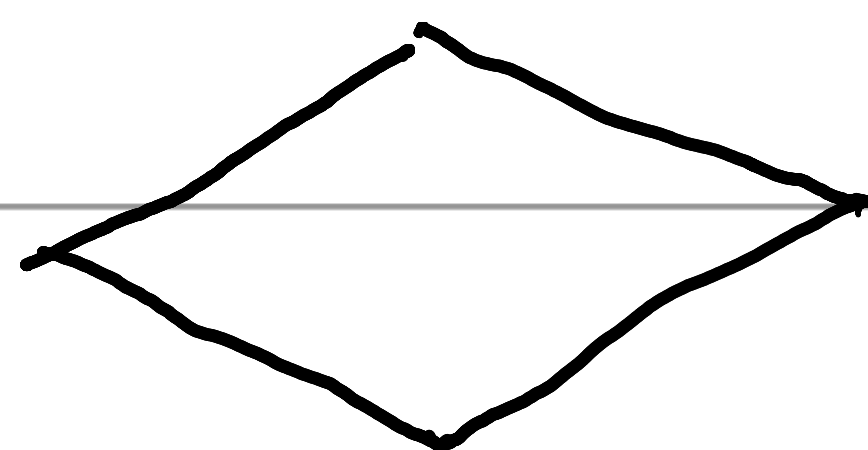
Square

All four sides are equal
The sides should be able to
be aligned horizontally
and vertically



Rectangle

Has four sides; Horizontal
and vertical; Opposite
sides are equal

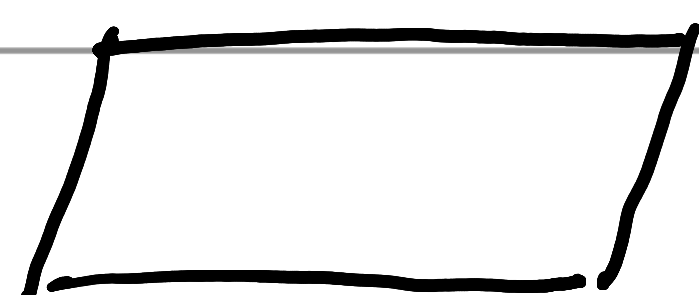


Rhombus

All four sides are equal;
Can be aligned in any
direction unlike square

Every square is a rhombus but

every rhombus is not a square



Parallelogram

4 sides; Opposite sides are
equal; Can be aligned
in any direction.

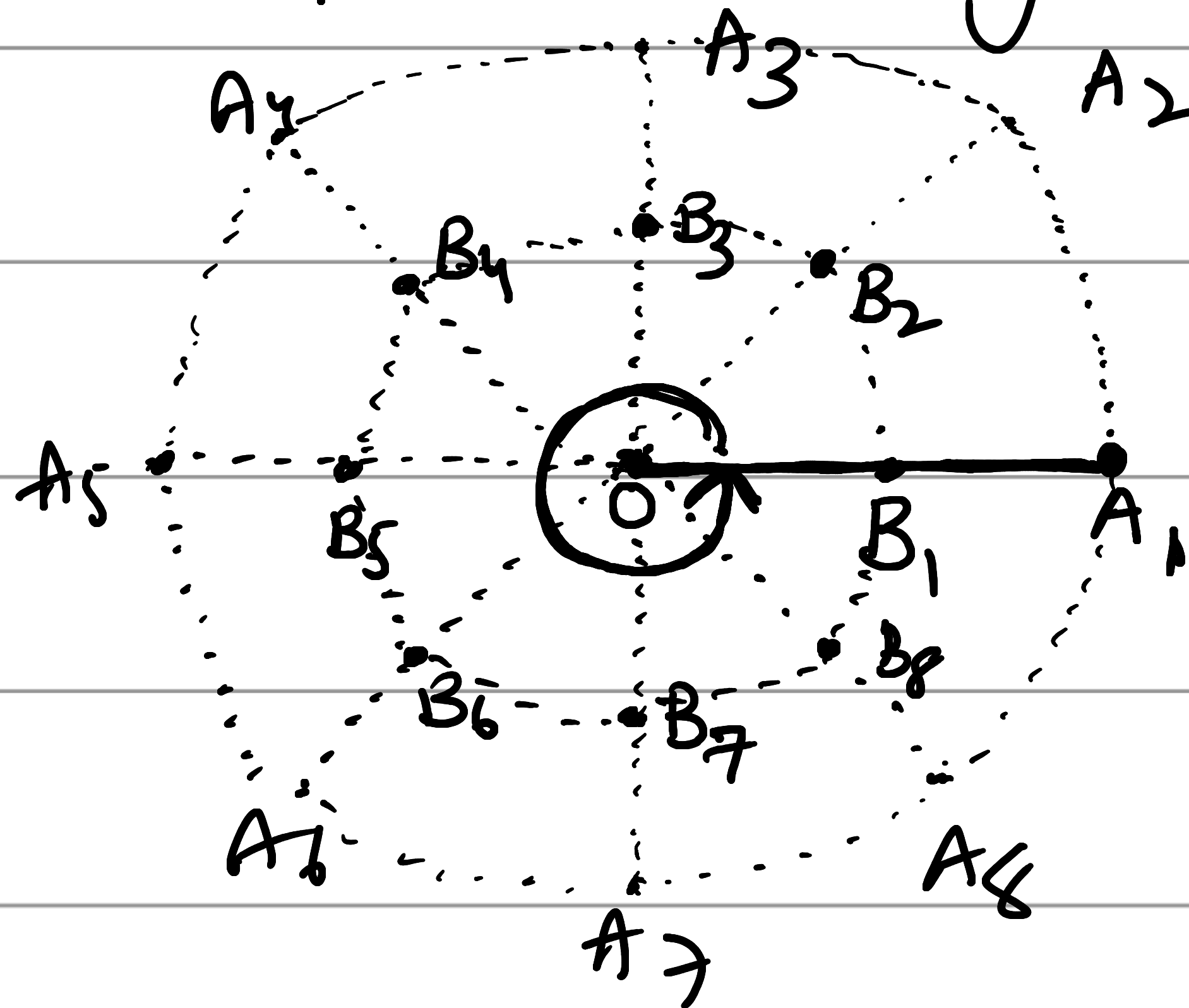
Every rectangle is a parallelogram but
not vice versa.



→ Triangle

- 3 sides
- and
- 3 angles

What is angle?



(Think 'O' is one end
of a stick fixed to the ground; 'A'
is the free end of the stick).

If we rotate OA_1 around
O, we get a circle.
Angle tells us how
much we have rotated.

Full rotation brings it back
to original position (OA_1).

If B_1 is the mid point of O and A_1 ,
then B_1 will also make a circle when A_1 makes
a circle around O.

Half rotation moves it to the opposite side (OA_5).

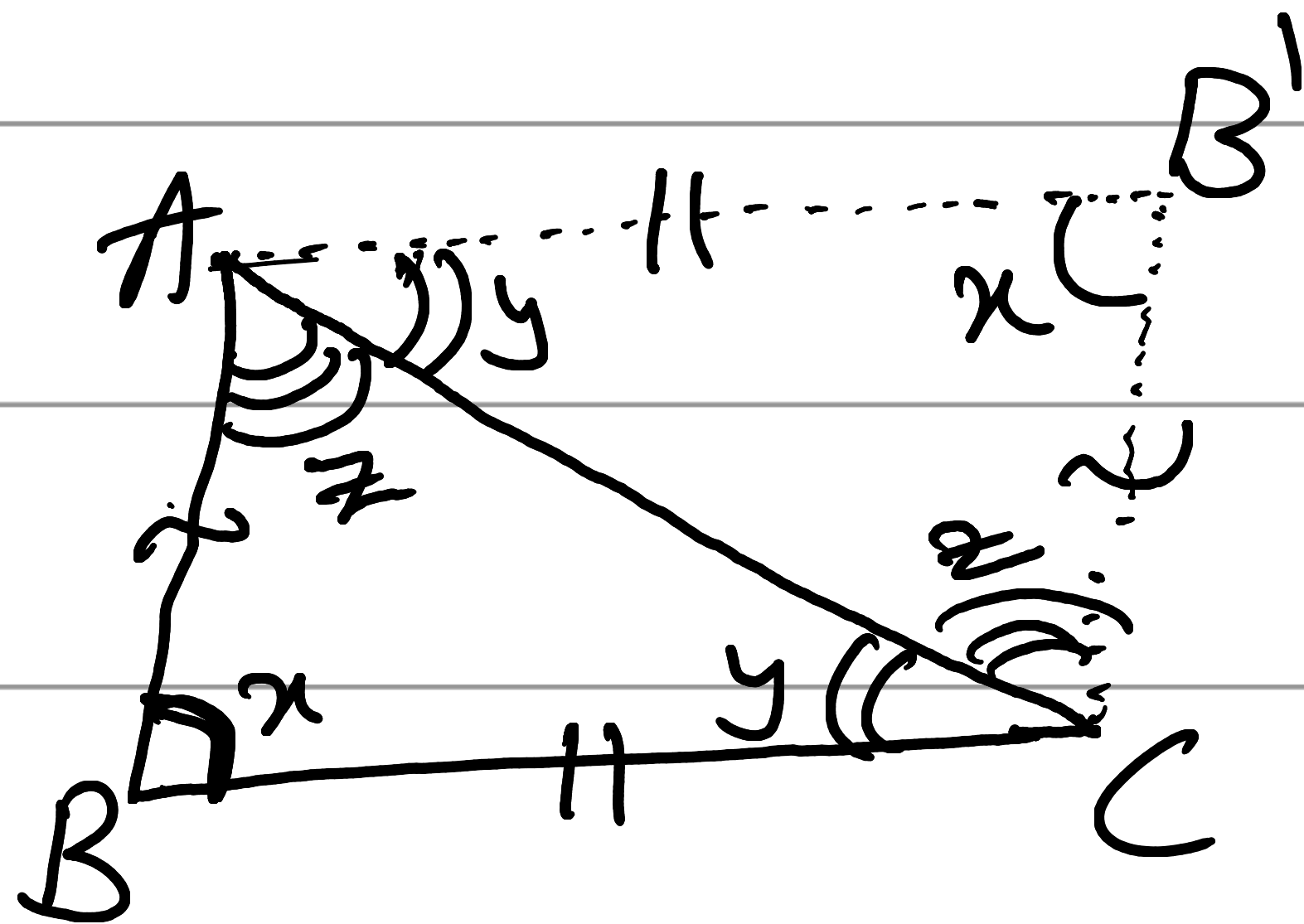
$\frac{1}{4}$ th rotation takes it from horizontal to vertical (OA_3)

So, in squares and rectangles all the angles are $\frac{1}{4}$ th of a full rotation.

→ See AI notes on Geometry - Introduction

Theorem: Sum of angles in a triangle is 180°

Proof: Let $\triangle ABC$ be any triangle.



Let B' be another point in the plane

Such that $\overline{AB'} = \overline{BC}$ & $\overline{AB} = \overline{B'C}$

So $ABCB'$ is parallelogram

Sum of angles in a parallelogram = $360^\circ \implies$ Why?

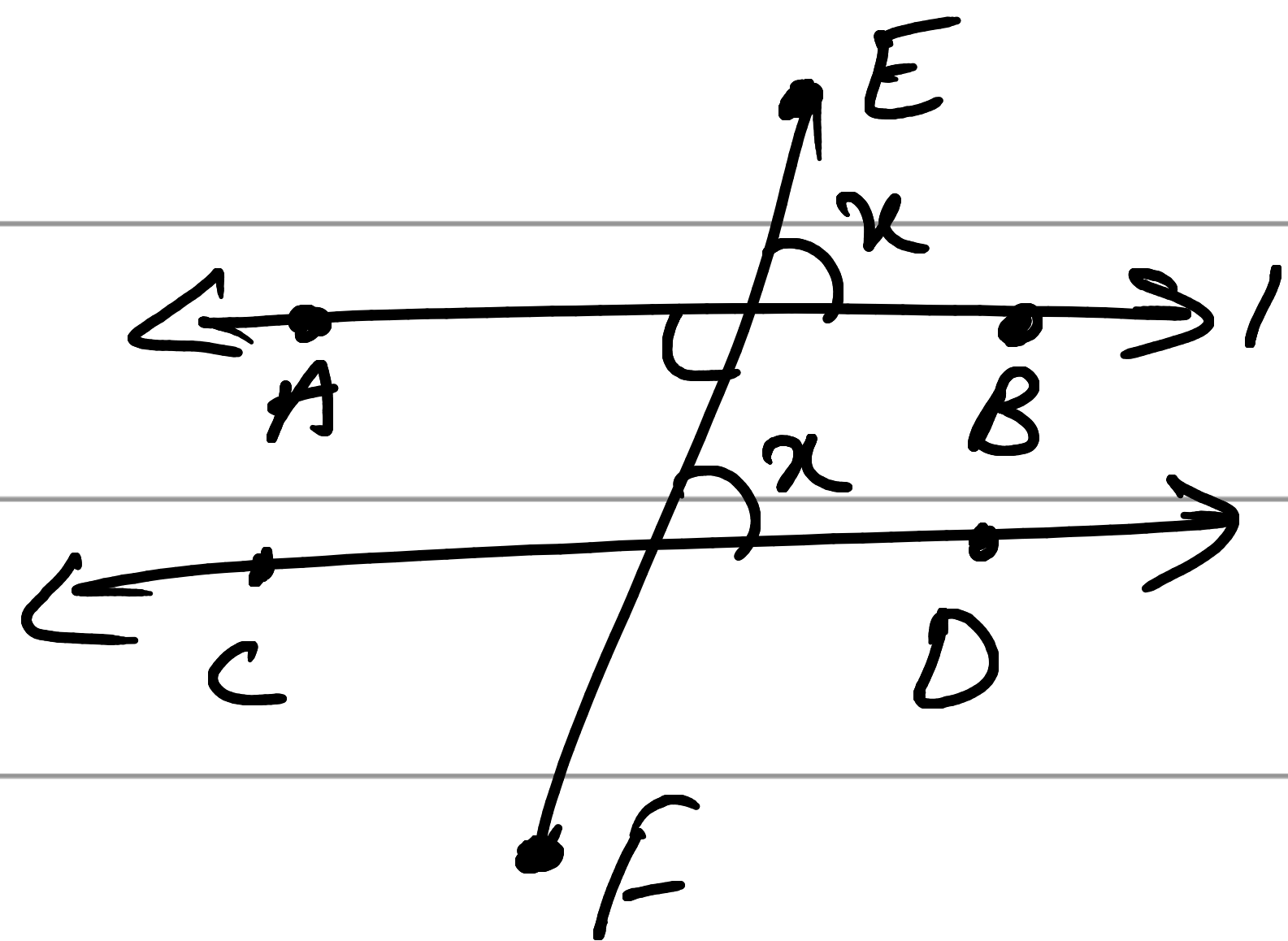
$$\angle B + \angle C + \angle B' + \angle A = x + (y+z) + x + (y+z) = 360^\circ$$

$$\implies 2(x+y+z) = 360^\circ$$

$$\implies x+y+z = 180^\circ \implies \text{Sum of angles of the triangle} = 180^\circ$$

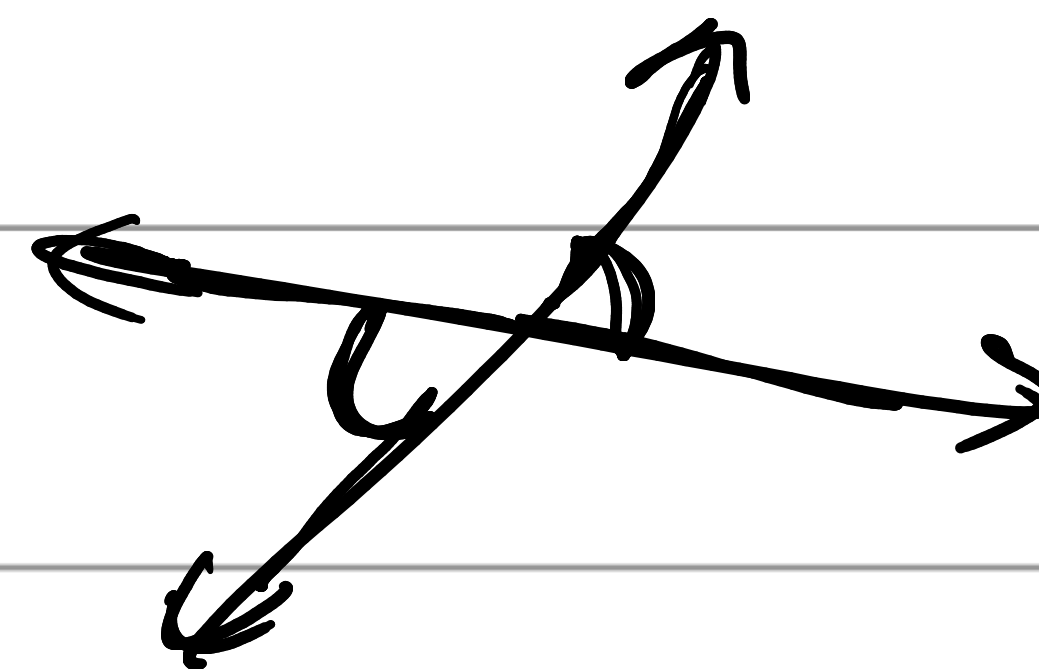
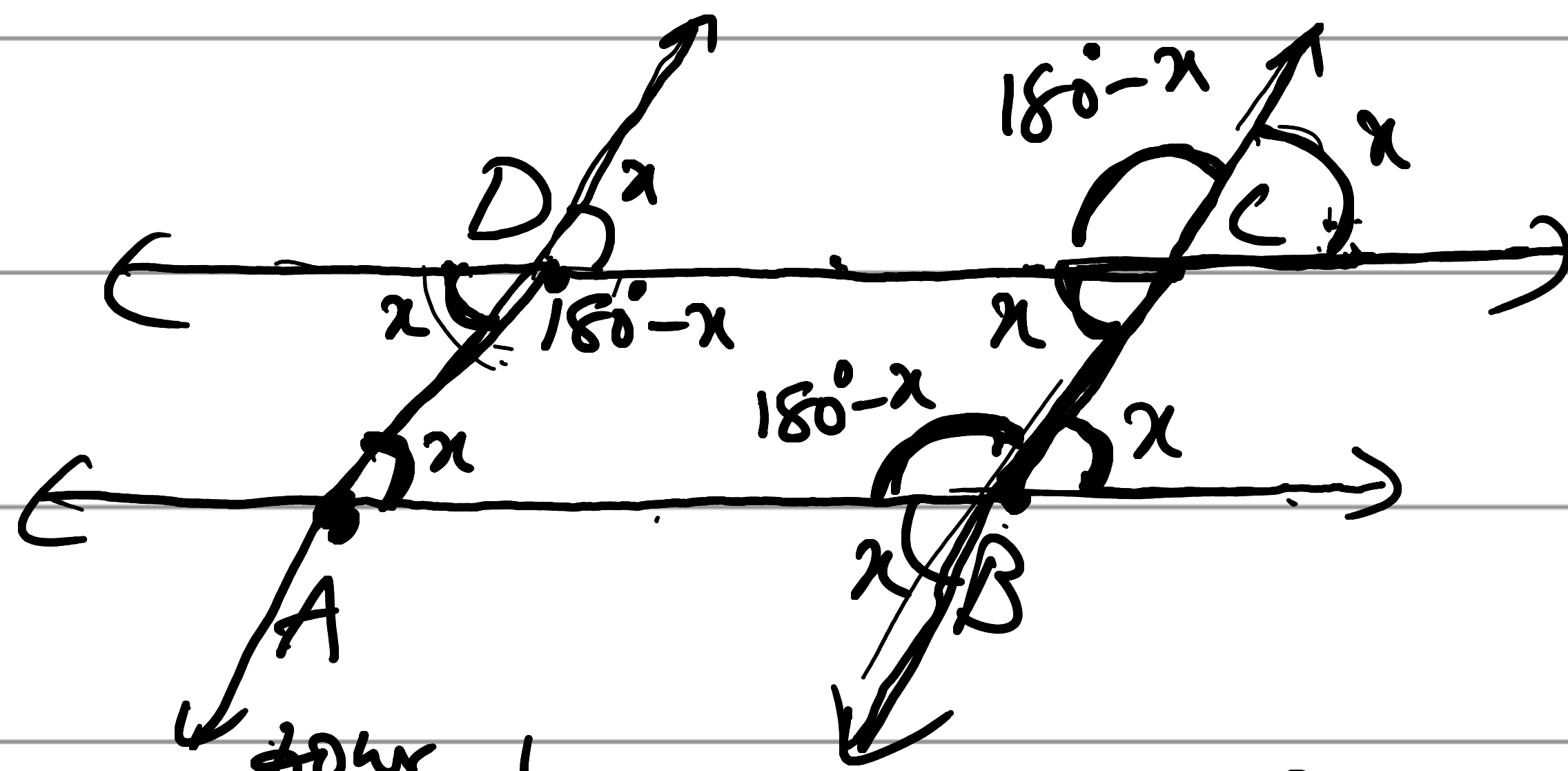
Theorem:- Sum of the angles in a parallelogram is 360° .

Proof



A line segment $\overline{EF}$ intersects two parallel lines $\overleftrightarrow{AB}$ and $\overleftrightarrow{CD}$ and makes the same angle ' x ' at both the parallel lines

Let take a parallelogram ABCD



$$\begin{aligned} \text{Sum of the angles in } ABCD &= x + (180^\circ - x) + x + (180^\circ - x) \\ \angle A + \angle B + \angle C + \angle D &= 360^\circ \end{aligned}$$

Axioms:

(1) $\overleftrightarrow{AB} \parallel \overleftrightarrow{CD}$
(parallel to)

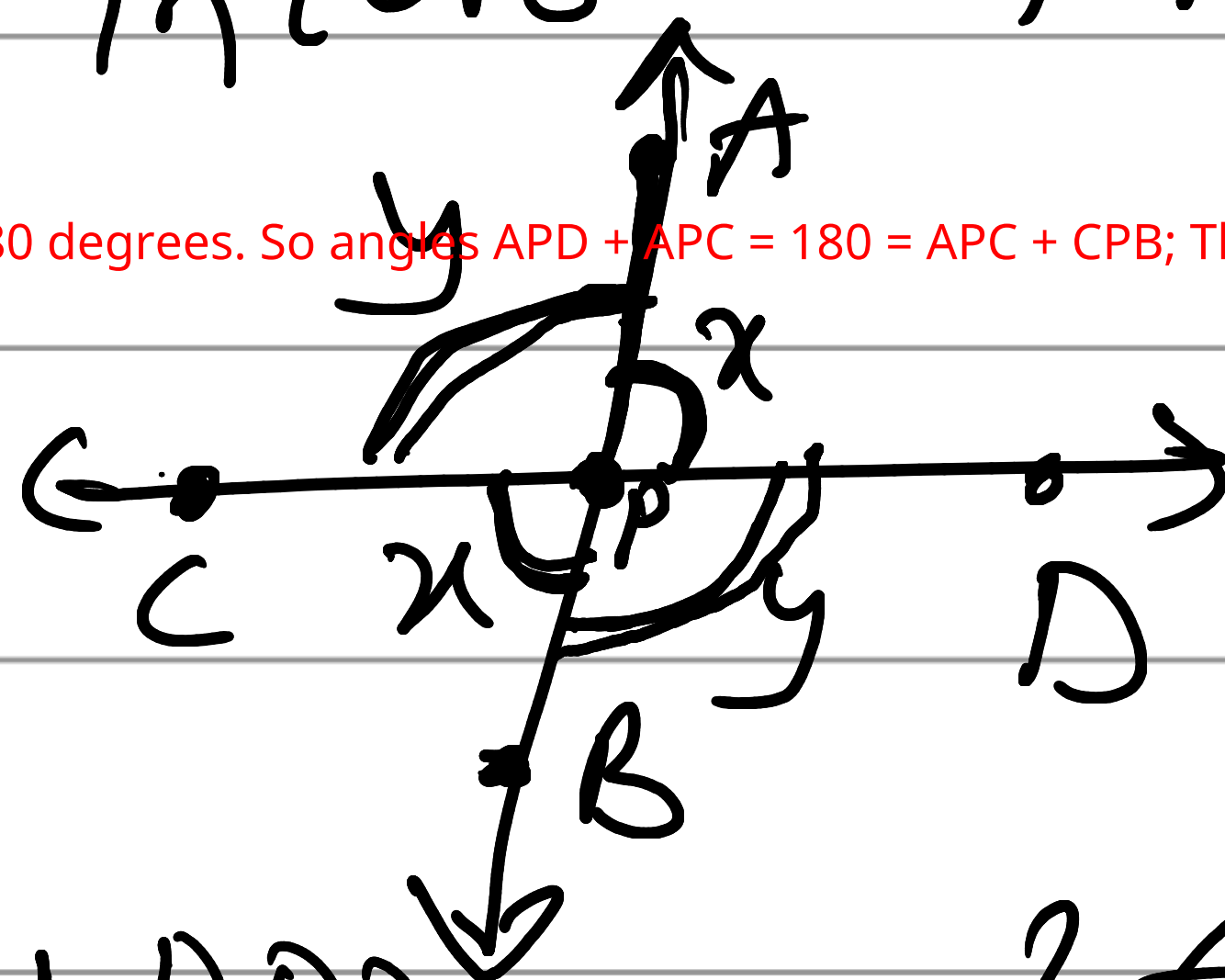


$\overleftrightarrow{EF}$ intersects $\overleftrightarrow{AB} \cap \overleftrightarrow{CD}$ at $P \cap P'$, respectively.

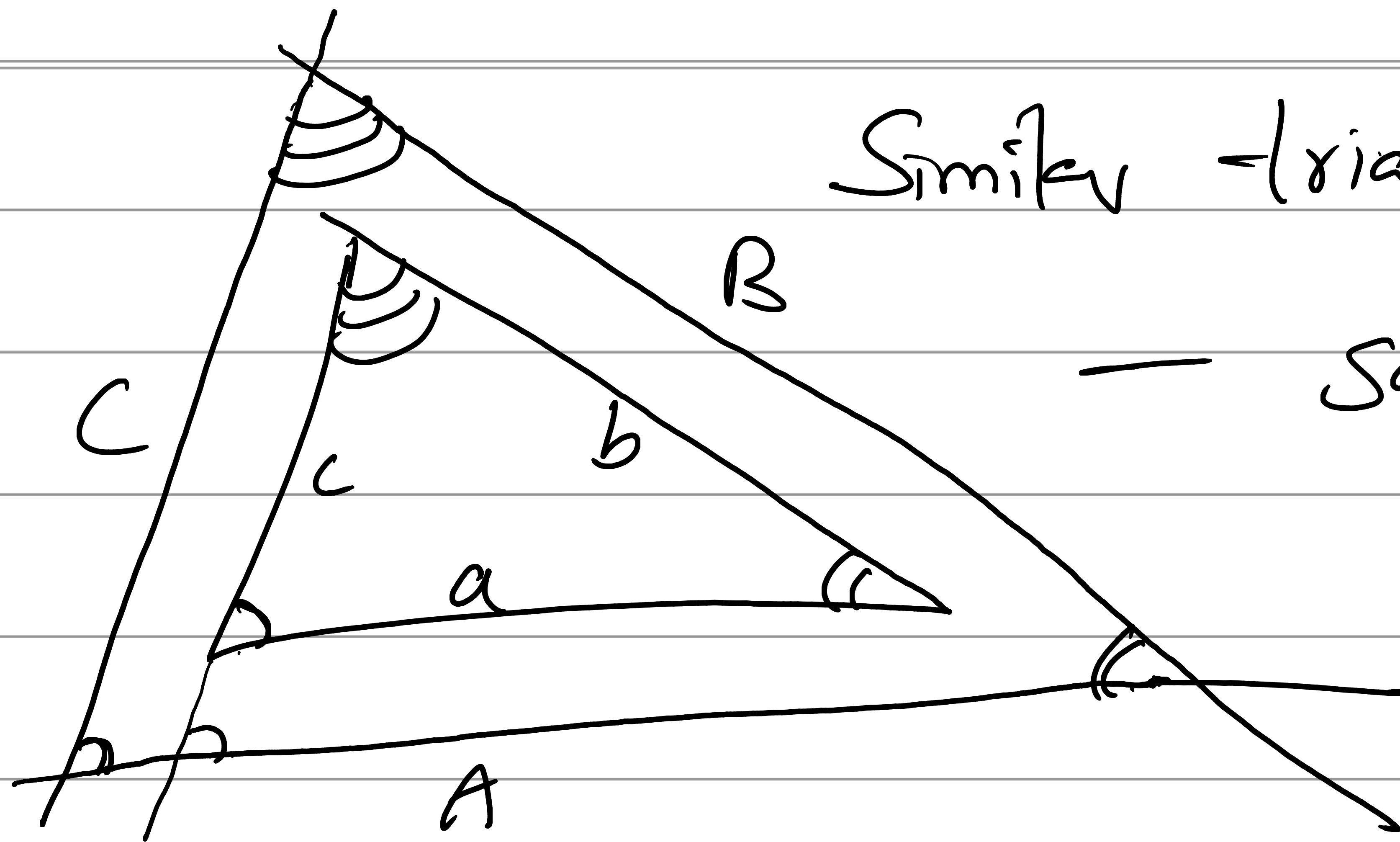
Then $\angle EPB = \angle EP'D$

(2) If $\overleftrightarrow{AB}$ intersects $\overleftrightarrow{CD}$, then at their point of intersection 'P', opposite angles are equal.

This can be derived from the fact that the angle made by a half rotation is 180 degrees. So angles $\angle APD + \angle BPC = 180 = \angle BPC + \angle CPB$; Therefore, $\angle APD = \angle CPB$. Opposite angles are equal. Similarly, $\angle CPA = \angle BPD$.



$$x + y = \angle APD + \angle BPC = \angle APB = \text{half rotation at 'P'} = 180^\circ.$$

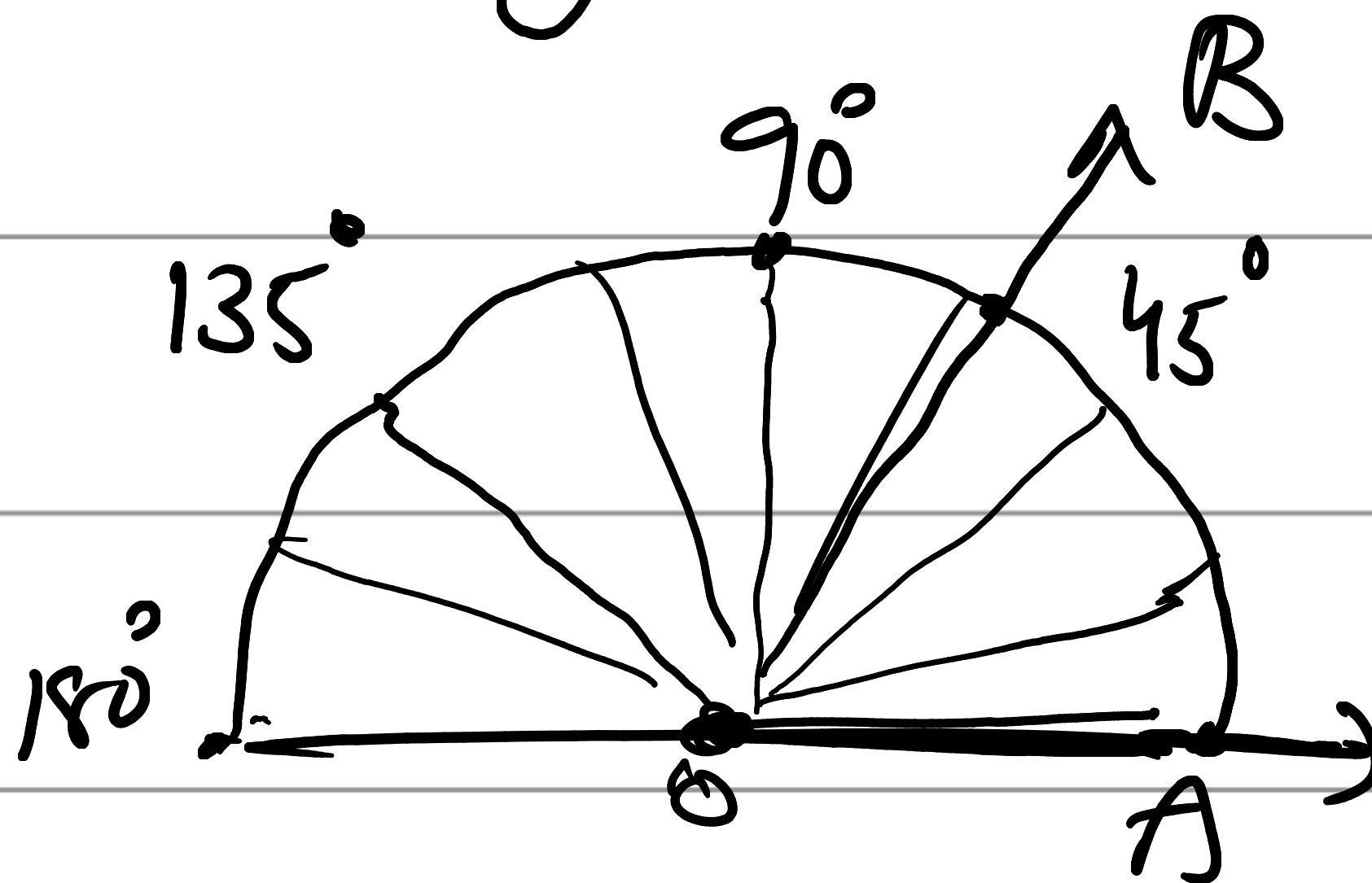
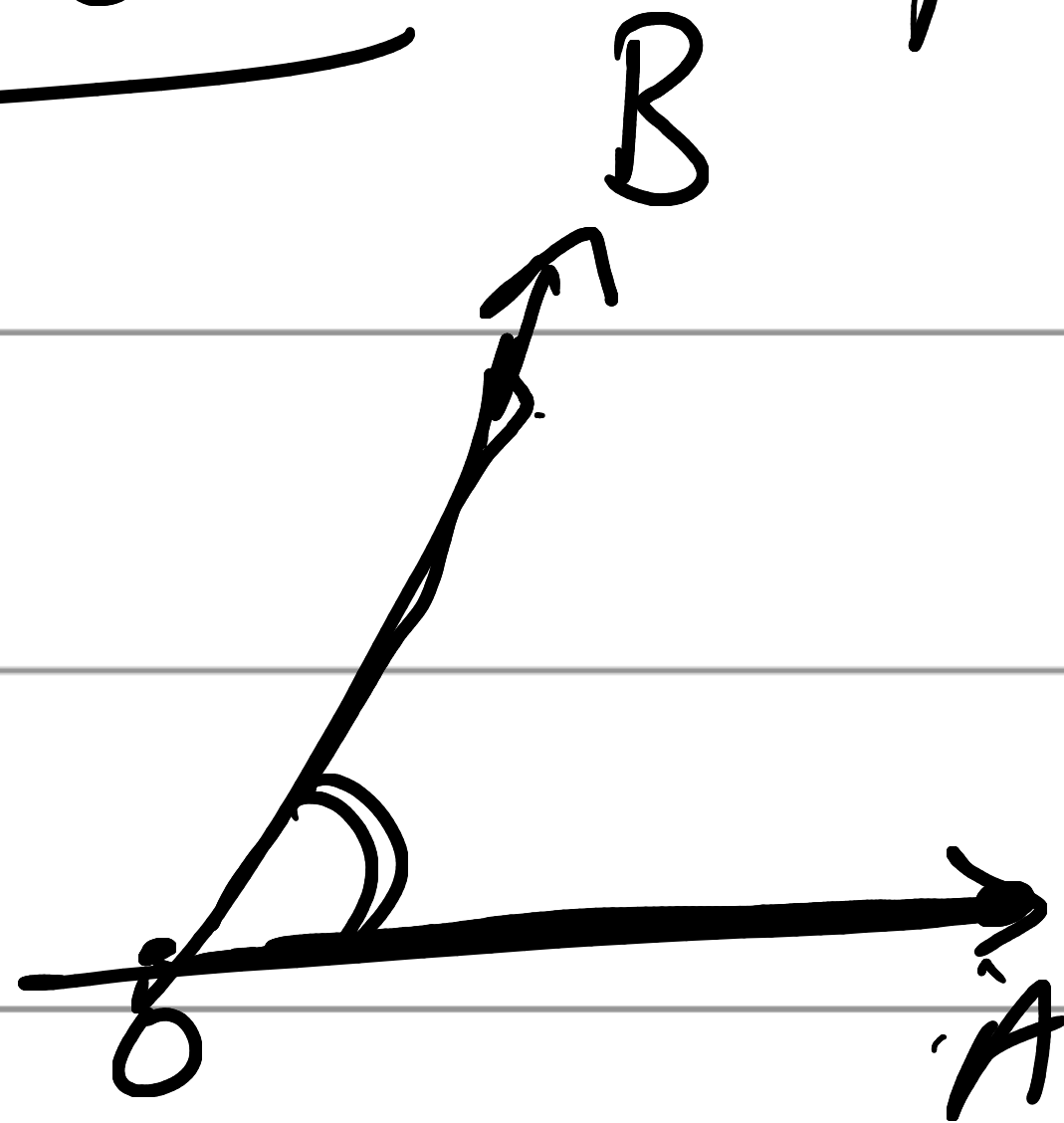


Similar triangles

— Same angles
but different
lengths
of sides.
(proportion)

$$\frac{a}{A} = \frac{b}{B} = \frac{c}{C}$$

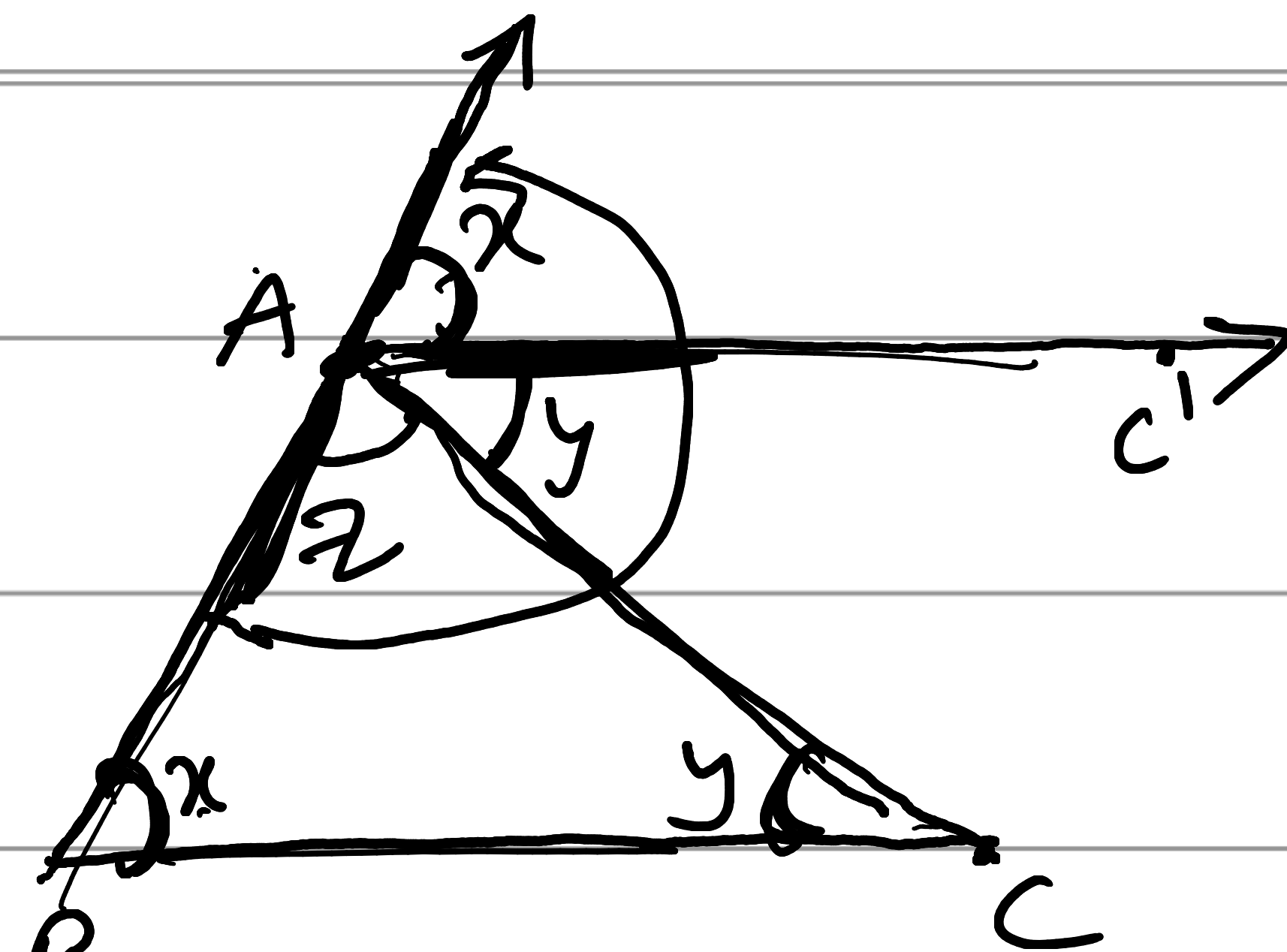
Protractor: Helps measure angles.



→ Sum of angles

in a

triangle:

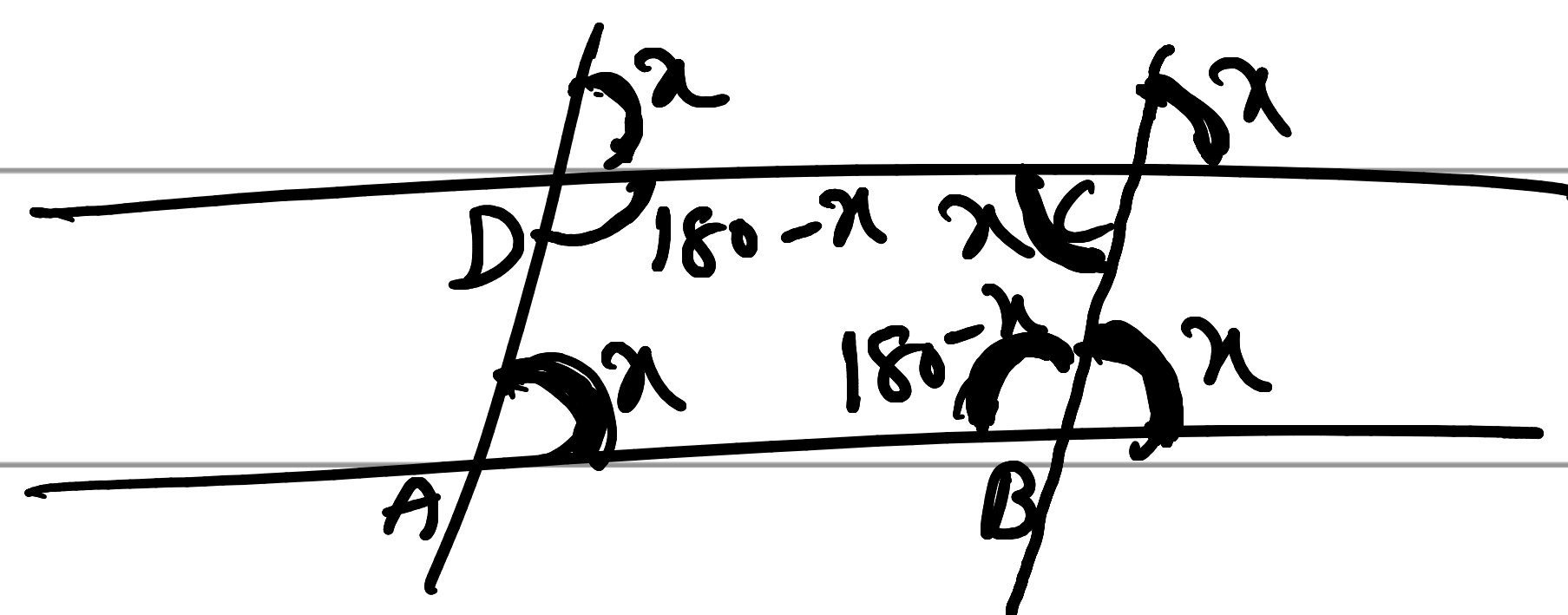


$$\vec{AC'} \parallel \vec{BC}$$

$$x + y + z = 180^\circ$$

(see AI description)

→ Sum of angles in a //gram



$$\begin{aligned} \text{Sum} &= \angle A + \angle B + \angle C + \angle D = x + (180^\circ - x) + x + (180^\circ - x) \\ &= 360^\circ \end{aligned}$$

HW: Read AI notes, class notes, solve problems, homework exercises.

HW: Buy a Geometry box

Todos for next class:

(1) Areas

(2) Circle — π

(3) Sets — see AI notes